Complete Sound Event Taxonomy and Characterization Guide

Cleaned, merged, and expanded version for audio annotation, sound-event detection, and dataset design

Version 2.0 - source-neutral labeling update, annotator clarity revision

Purpose
This guide converts the original brainstorm-style list into a consistent taxonomy. It removes duplicate sections, standardizes labels, adds acoustic characterization, and includes annotation rules for reliable human or machine-learning use.

Document map

- Section 1 defines the recommended structure and the acoustic descriptors.
- Section 2 gives a compact category overview.
- Section 3 provides the full sound-event taxonomy with examples, acoustic profiles, and annotation notes.
- Section 4 lists common confusable sound pairs.
- Section 5 gives annotation guidelines and quality-control rules.

1. Recommended taxonomy structure

Recommended label hierarchy: Top category > subcategory > canonical acoustic label > optional source/context > acoustic descriptors > annotation notes.

Source-neutral canonical label rule: Do not encode the source in the canonical label unless the source is clearly identifiable from audio alone. If the source is uncertain, use a source-neutral acoustic label such as rustling sound, hissing sound, clicking sound, or impact sound, and store the suspected source separately as optional context.

Field	Recommended use	Example
Top category	Broad source family used for navigation and reporting.	Human, animal, mechanical, liquid, environment
Subcategory	More specific source or mechanism.	Speech, breathing, impact, friction, airflow
Canonical label	The standardized acoustic label used in the dataset. It should usually describe what is heard, not the guessed source.	Coughing, rustling sound, hissing sound, rain pattering
Description	Plain-language definition of the event.	Sharp expulsion of air to clear airways
Example	Concrete real-world case for annotators.	A dry cough in a dusty room
Acoustic profile	Observable audio traits, independent of visual context.	Impulsive, broadband, repeated bursts
Annotation note	How to distinguish this label from similar labels.	Coughing is stronger than throat clearing
Optional source/context	Use only when the source is clear from audio or verified metadata. Do not force it from visual context alone.	paper, leaves, tire, door, fan, unknown

Acoustic characterization descriptors

Descriptor	Allowed values or examples	How to use it
Duration	Impulsive (<50 ms), short (50–500 ms), medium (0.5–3 s), sustained (>3 s)	Clap is impulsive; rain is sustained
Temporal pattern	Single event, repeated, rhythmic, intermittent, continuous	Typing is repeated; alarm is repetitive; hum is continuous
Pitch range	Low, mid, high, broadband	Thunder is low; squeak is high; crash is broadband
Spectral type	Tonal, harmonic, noisy, broadband, narrowband	Doorbell is tonal; cough is broadband and noisy
Envelope	Sharp attack, slow attack, decaying tail, steady state	Bang has sharp attack; ringing has decaying tail
Source mechanism	Vocalization, airflow, impact, friction, vibration, flow	Whistle is airflow; scrape is friction; knock is impact
Spatial context	Near field, far field, indoor, outdoor, moving source	Car pass-by changes level and spectrum over time
Annotation priority	Dominant, foreground, background, uncertain	Use dominant label when one event clearly masks others

2. Category overview

Top category	Scope	Representative labels
Human vocal and speech	Vocal tract sounds produced intentionally or emotionally	Talking, whispering, singing, laughter
Human respiratory and involuntary	Breath, airway, and reflex sounds	Coughing, sneezing, snoring, wheezing
Human digestive, body, and contact	Mouth, stomach, hand, foot, and body contact sounds	Chewing, burping, footsteps, clapping
Baby and infant	Age-specific pre-speech and distress sounds	Cooing, babbling, wailing
Social and crowd	Group vocal and collective interaction sounds	Cheering, booing, applause
Animal	Animal vocalizations and body generated sounds	Barking, purring, chirping, buzzing
Nature and environment	Weather, geophysical, vegetation, fire, and outdoor ambience	Thunder, rain, wind, wave crashing, fire crackle, rustling sound
Liquid and fluid	Water, liquid movement, gas in liquid, and liquid impact	Dripping, splashing, gurgling, spraying
Impact and collision	Short events caused by contact, breaking, or collision	Bang, crash, thud, pop
Friction and texture	Rubbing, scraping, crinkling, and surface interaction	Scratching, squeaking, rustling, crinkling, rubbing
Metallic and resonant	Metal or glass contact and ringing resonances	Clanging, clinking, jingling
Mechanical, appliance, and industrial	Motors, rotating systems, engines, and machine vibration	Whirring, hum, rattle, roar, whooshing, grinding
Air and pressure	Gas flow, pressurized release, airflow through gaps	Hiss, whoosh, puff, whistle, pressure release
Electronic, alert, and interface	Digital, electrical, alarm, and user interface sounds	Beep, alarm, notification tone, tapping, clicking
Music and tonal	Intentional tones, instruments, and rhythmic musical sounds	Chime, strum, drum beat, twang
Food preparation and cooking	Kitchen preparation plus heated food and liquid sounds	Chopping, frying, boiling, kneading, popping, crackling
Transportation and vehicles	Cars, trains, aircraft, motorcycles, horns, and road sounds	Pass-by, horn, rumble, siren, tire squeal, rotor sound

Top category	Scope	Representative labels
Construction and tools	Human-operated tools and worksite events	Hammering, drilling, sawing, sanding, ratcheting, dropping metal objects

3. Full sound-event taxonomy

Each table uses standardized labels, avoids repeated categories, and favors source-neutral canonical labels when the sound source is uncertain. The acoustic profile column is intended to make the document useful for annotation, audio ML, and sound-event detection.

Human vocal and speech sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Speech	Talking or chatting	Continuous conversational speech from one or more people.	Two colleagues discussing a project.	Medium to long; voiced; harmonic plus consonant noise.	Separate from crowd murmur when individual speech is intelligible.
Speech	Whispering	Soft breathy speech with little or no vocal fold vibration.	Whispering in a library.	Low level; high breath noise; weak pitch.	Do not label as quiet talking if voicing is absent or minimal.
Speech	Shouting	Loud projected speech with strong vocal energy.	Calling to someone across a street.	High level; strong harmonics; sharp syllable envelope.	Can overlap with screaming; use shouting when words are speech-like.
Speech	Singing	Melodic vocalization with sustained pitch and rhythm.	Singing along to a song in a car.	Tonal; harmonic; pitch stable or melodic.	Do not merge with humming when mouth is open and lyrics may be present.
Speech	Humming	Closed-mouth tonal vocalization.	Humming a tune while working.	Tonal; nasal; sustained harmonic structure.	Often confused with mechanical hum; use context and pitch variation.
Expression	Laughing	Vocal expression of amusement, from chuckles to loud laughter.	A person laughing at a joke.	Repeated voiced bursts; variable pitch; rhythmic.	Use giggling for light repeated childlike laughter.
Expression	Giggling	Light, high-pitched repeated laughter.	Children giggling during a game.	Short repeated bursts; high pitch; low to medium level.	Subtype of laughter; useful when fine labels are needed.
Expression	Crying or sobbing	Vocal expression of sadness or distress, often with sobs.	A person crying during an emotional scene.	Irregular voiced bursts; breathy inhalations; variable intensity.	For infants, use baby crying when age/context is clear.
Expression	Screaming	Very loud high-energy vocalization from fear, pain, or excitement.	Screaming on a roller coaster.	High level; high pitch; strong nonlinear voice quality.	Use shouting if speech-like words dominate.
Expression	Sighing	Long audible exhale expressing relief, fatigue, or frustration.	Sighing after a long workday.	Short to medium; breath noise; decaying envelope.	May be confused with breathing; label only if clearly expressive.
Expression	Moaning or groaning	Low prolonged vocal sound expressing discomfort, effort, or pleasure.	Groaning while lifting a heavy object.	Low pitch; sustained; voiced; slow envelope.	Separate from structural groaning by source context.
Mouth generated	Whistling	High-pitched tone made by forcing air through lips or teeth.	Whistling a melody while walking.	Tonal; high frequency; narrowband; sustained or melodic.	Air whistle from wind belongs under air and pressure if non-human.
Exertion	Grunting	Short forceful voiced sound made during physical effort.	Grunting while lifting a heavy weight.	Short; low pitch; voiced; sharp onset.	Distinct from moaning or groaning, which are longer and less effortful. Use during fitness or lifting context.

Human respiratory and involuntary sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Reflex	Sneezing	Sudden explosive burst of air through nose and mouth.	Sneezing during allergy season.	Impulsive; broadband; strong onset; short tail.	Usually followed by nasal/breath sounds; do not label as cough.
Reflex	Coughing	Sharp expulsion of air to clear throat or airways.	A dry cough in a dusty room.	Impulsive repeated bursts; broadband; chest/throat resonance.	Confusable with throat clearing; cough is stronger and more explosive.
Deliberate airway	Throat clearing	Short guttural sound used to clear mucus or prepare to speak.	Clearing throat before a presentation.	Short; low to mid frequency; rough vocal quality.	Less explosive than coughing.
Sleep/breathing	Snoring	Rhythmic vibrating sound during sleep from relaxed airway tissues.	Loud nasal snoring at night.	Low to mid frequency; periodic; voiced/vibratory.	Use breathing if no rough vibration is present.
Breathing	Wheezing	High-pitched strained breathing sound caused by narrow airflow.	Wheezing during heavy exertion.	High pitch; narrowband; often sustained during inhale or exhale.	Can be medical context; annotate acoustically without diagnosis.
Breathing	Gasping	Sudden sharp intake of breath.	Gasping after a surprise.	Short; noisy inhale; fast attack.	Different from sighing, which is usually a long exhale.
Breathing	Yawning	Deep inhalation with open mouth, often expressing fatigue.	Yawning during a meeting.	Medium duration; breath noise plus voiced onset or tail.	Can include vocalization; keep as yawning if mouth/breath gesture dominates.
Reflex	Hiccupping	Involuntary diaphragm spasm producing a short hic sound.	Hiccups after eating too quickly.	Short repeated pulses; vocal/glottal onset.	Usually periodic but sparse.
Breathing	Panting	Rapid shallow breathing after exertion or heat.	Panting after climbing stairs.	Repeated breath bursts; low to medium level; periodic.	For animals, use animal panting only if source taxonomy supports it.
Nasal	Sniffling	Repeated short nasal inhalations.	Sniffling with a runny nose.	Short nasal inhales; low level; repeated.	May co-occur with crying; label dominant event.
Breathing	Breathing	Audible inhale and exhale without another stronger respiratory event.	Slow breathing close to a microphone.	Soft broadband airflow; periodic; low level.	Use only when breathing is audible and relevant.

Human digestive, body, and contact sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Digestive	Burping or belching	Release of gas from the stomach through the mouth.	Burping after a carbonated drink.	Short to medium; low pitch; voiced or noisy.	Can be socially sensitive; keep neutral label wording.
Digestive	Vomiting or gagging	Retching or forceful expulsion sounds from throat/stomach.	Gagging after a bad taste.	Irregular; wet/throat noise; strong low-frequency components.	Use only when acoustically clear; otherwise mark uncertain.
Digestive	Stomach growling	Rumbling sound from digestive activity.	Stomach gurgling in a quiet room.	Low frequency; irregular; soft rumble/gurgle.	Often confused with external liquid gurgling; source context helps.
Mouth/eating	Swallowing or gulping	Audible throat movement when ingesting liquid or food.	Gulping water after exercise.	Short wet click; low level; close microphone.	May be hidden by slurping or chewing.
Mouth/eating	Chewing or crunching	Teeth breaking down food.	Crunching on a crisp apple.	Repeated short impacts; broadband; sometimes wet texture.	Crunching is sharper than chewing soft food.
Mouth/eating	Slurping	Noisy suction of liquid into the mouth.	Slurping hot soup.	Sustained noisy suction; wet; mid/high airflow.	Distinguish from straw sucking or drinking if needed.
Mouth/eating	Lip smacking	Short wet mouth contact sounds.	Lip smacking after tasting food.	Short clicks; wet texture; close mic.	May be part of eating; use fine label only when prominent.
Body contact	Footsteps	Repeated foot-ground impacts from walking or running.	Someone walking on a wooden floor.	Rhythmic impacts; surface-dependent spectrum.	Use surface-specific labels only if taxonomy supports them.
Body contact	Hand clapping	Sharp sound from striking palms together.	Audience member clapping once.	Impulsive; broadband; fast attack; short decay.	Collective applause belongs under social sounds.
Body contact	Finger snapping	Short sharp sound from finger friction and release.	Snapping fingers to a beat.	Impulsive; high frequency; short decay.	Can resemble small click; source context helps.

Baby and infant sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Pre-speech	Cooing	Soft vowel-like infant vocalization expressing comfort.	A baby cooing at a parent.	Soft; voiced; smooth pitch; low intensity.	Use only when infant source is clear.
Pre-speech	Babbling	Repetitive consonant-vowel combinations in pre-speech.	A baby saying ba-ba-ba.	Repeated syllables; voiced; rhythmic.	Different from adult speech because words are not formed.
Distress	Baby crying or wailing	Loud sustained infant distress vocalization.	A hungry baby crying at night.	High pitch; strong modulation; sustained or repeated.	Keep separate from adult crying when age/source is clear.
Wet vocal	Baby gurgling	Bubbly wet infant vocalization from saliva and air.	A baby gurgling after feeding.	Wet texture; soft bubbles; voiced components.	Can overlap with liquid gurgling; source context matters.
Distress	Fussing or whimpering	Low-level intermittent infant distress sounds.	A tired baby fussing before sleep.	Soft repeated whines; variable pitch.	Less intense than baby crying or wailing.
Excited vocal	Baby squealing	High-pitched excited infant vocalization.	A baby squealing during peekaboo.	Very high pitch; sharp onset; sustained or repeated.	Can resemble scream; infant context defines label.

Social and crowd sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Audience	Applause	Sustained collective clapping from multiple people.	Standing ovation in a theater.	Many overlapping impulsive claps; dense texture.	Use hand clapping for a single person or isolated clap.
Audience	Cheering	Loud enthusiastic vocal celebration by one or many people.	Crowd cheering after a goal.	High energy; many voices; wideband; sustained.	May co-occur with applause; label dominant component.
Audience	Booing	Vocal disapproval from a crowd.	Booing a bad call by a referee.	Sustained vowel-like low/mid vocal energy.	Can sound like crowd murmur; use when disapproval is clear.
Group vocal	Chanting	Rhythmic repeated group vocalization.	Fans chanting a team name.	Rhythmic; repeated phrases; many voices.	Separate from singing if speech rhythm dominates.
Crowd bed	Crowd murmur	Diffuse background voices without clear individual speech.	Restaurant or lobby ambience.	Continuous; many voices; low intelligibility.	Use talking if a foreground speaker is intelligible.
Crowd expression	Crowd laughter	Multiple people laughing together.	Audience laughing at a comedy show.	Overlapping laughter bursts; broad dynamic range.	Use laughing for a single or foreground person.

Animal sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Dog	Barking	Short loud dog vocal bursts.	A dog barking at a door.	Impulsive voiced bursts; mid frequency; repeated.	Growling is lower and more sustained.
Dog	Growling	Low rumbling animal threat sound.	A dog growling at a stranger.	Low frequency; rough; sustained or pulsed.	Can resemble engine rumble; source context needed.

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Cat	Meowing	Cat vocal call with changing pitch.	A cat meowing for food.	Tonal; pitch glide; short to medium duration.	Separate from purring and hissing.
Cat	Purring	Soft continuous vibration from a cat.	A cat purring while being petted.	Low level; periodic vibration; low frequency.	Can be confused with mechanical hum in close recordings.
Cat/reptile	Hissing	Sharp sustained warning sound from animal mouth.	A cat hissing at a stranger.	Broadband high-frequency noise; sustained.	Air leak hiss is non-animal; source/context matters.
Bird	Chirping or tweeting	Short high-pitched bird calls.	Birds chirping at sunrise.	High frequency; tonal; repeated.	Use screeching for harsh loud bird cries.
Bird	Screeching	Loud harsh high-pitched bird or animal cry.	A parrot screeching.	High level; high pitch; rough; broadband elements.	Can overlap with human screaming; source context matters.
Insect	Buzzing insect	Rapid wing vibration from an insect.	A mosquito buzzing near an ear.	Narrowband buzz; high frequency; moving source.	Mechanical buzzing is steadier and non-biological.
Amphibian	Croaking	Low rough repetitive frog or toad call.	Frogs croaking at a pond.	Low to mid frequency; repeated pulses.	Often occurs in outdoor night ambience.
Wild animal	Howling	Long mournful animal vocalization.	A coyote howling at dusk.	Sustained; pitch glide; long decay.	Can resemble wind howl; source context helps.
Reptile	Rattlesnake rattling	Rapid dry rattle from tail movement.	A rattlesnake shaking its tail.	Rapid repeated clicks; dry texture; high frequency.	Can resemble keys or small object rattle.
Small mammal	Chittering	Rapid high-pitched chatter from small animals.	Squirrels chittering in a tree.	Fast repeated high-pitched pulses.	Can be confused with bird chatter.
Bird	Rooster crowing	Distinctive loud rooster call.	A rooster crowing at dawn.	Tonal; strong pitch contour; medium duration.	Fine animal label when source is known.
Livestock	Horse neighing	High-energy horse vocalization.	A horse neighing in a stable.	Tonal with noisy onset; pitch variation.	Use broader animal vocalization if source unclear.
Livestock	Cow mooing	Low sustained cattle vocalization.	A cow mooing in a field.	Low pitch; harmonic; sustained.	Can resemble foghorn-like tones in poor context.

Nature and environmental sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Weather	Thunder	Deep rumbling boom caused by lightning.	Thunderclap during a summer storm.	Low frequency; impulsive onset with long rumble tail.	Do not confuse with explosion if weather context is present.
Weather	Wind howling	Sustained moaning sound of strong air currents.	Wind through a narrow alley.	Broadband airflow with tonal resonances; sustained.	Can resemble animal howling; wind lacks vocal harmonics.
Weather	Rain pattering	Many raindrops tapping surfaces.	Rain on a window or roof.	Dense fine impacts; broadband; steady bed.	Heavy rain may mask other events.
Weather	Hail	Hard pelting sound of ice striking surfaces.	Hailstones hitting a car roof.	Sharper and harder impacts than rain; irregular.	Can resemble small object drops on metal.
Geophysical	Avalanche rumble	Deep growing roar of snow and debris moving downhill.	Snow cascading down a mountain.	Low frequency; increasing intensity; sustained.	Rare label; use only if context is reliable.
Geophysical	Earthquake rumble	Low-frequency ground-shaking vibration.	Deep rumble during ground movement.	Very low frequency; long rumble; structural rattles may co-occur.	Often inferred from context; avoid over-labeling.
Water body	Wave crashing	Powerful rolling ocean or lake water breaking on shore.	Surf pounding a rocky coastline.	Low to high broadband rush; repeated swells.	May include splash and foam hiss.
Fire/vegetation	Fire crackling	Irregular snapping of burning material.	Campfire crackling at night.	Short irregular pops and crackles; high frequency.	Can resemble cooking crackle; context matters.
Vegetation	Rustling sound	Soft swishing from leaves or light vegetation moving.	Dry leaves blowing across a sidewalk.	Soft broadband; irregular; low to mid level.	Use this source-neutral label when the source is unclear. Add paper, leaves, clothing, or unknown as optional source/context only if supported.
Water body	Flowing water	Continuous water movement over rocks or terrain.	A stream flowing over pebbles.	Continuous broadband with small gurgles.	Use source-neutral water-flow label unless stream, pipe, or fountain source is clearly known.
Water body	Continuous water roar	Continuous heavy falling water.	A waterfall in a canyon.	Loud broadband noise; sustained; low to high energy.	Use this when the sound is continuous water noise; store waterfall as optional context only if clear.
Water body	Surf or rolling waves	Repeated distant waves and foam wash.	Distant surf along a beach.	Cyclic broadband wash; medium to long duration.	Use wave crashing for individual foreground impacts; store ocean/coast context separately when needed.
Ice/natural material	Cracking sound	Sharp fracture sounds from ice stress.	Lake ice cracking in winter.	Impulsive crack; high frequency; sometimes resonant tail.	Use source-neutral cracking if ice, branch, plastic, or other source is uncertain.

Liquid and fluid sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Impact	Splashing	Object or body displacing liquid suddenly.	Jumping into a puddle.	Impulsive plus wet broadband tail.	Use wave crash for large natural water bodies.
Drops	Dripping	Single drops falling at intervals.	Leaky faucet at night.	Repeated isolated clicks; wet resonant tail.	Can resemble ticking in reverberant rooms.
Flow	Gurgling	Irregular bubbling or draining flow.	Water going down a drain.	Wet bubbling; intermittent low/mid frequencies.	May resemble stomach growling; source context helps.
Flow	Pouring	Steady stream of liquid entering a container.	Pouring coffee from a carafe.	Continuous wet noise; level changes with flow.	Use gushing when flow is forceful and sudden.
Container motion	Sloshing	Heavy liquid shifting inside a container.	Water sloshing in a half-full bottle.	Low-frequency wet impacts; irregular movement.	Often occurs with object handling noise.
Gas in liquid	Bubbling	Repeated gas bubbles escaping through liquid.	A fish tank aerator.	Small repeated pops; wet; often periodic or dense.	Boiling is heat-driven bubbling in cooking context.
Pressure	Spraying	Fine mist or liquid expelled under pressure.	Spraying a garden hose.	Noisy broadband jet; sustained; high-frequency hiss.	Can resemble air hiss if liquid impact is weak.
Flow	Gushing	Sudden forceful rush of liquid.	Water bursting from a hydrant.	High energy; broadband; strong onset.	Use pouring for controlled steady flow.
Small flow	Trickling	Light continuous small stream of liquid.	A gentle brook over pebbles.	Soft high-frequency water texture; continuous.	Can be background ambience.
Heat/liquid	Sizzling sound	Hissing from moisture contacting a hot surface.	Water droplets on a hot pan.	Continuous high-frequency hiss plus small pops.	Cooking sizzle may include food and oil crackle.
Plumbing	Toilet flushing	Rushing and gurgling water from a toilet flush cycle, ending with tank refill.	A toilet flushing in a bathroom.	Sustained turbulent water rush; gurgle; refill hiss tail.	Distinct from sink or shower by the surge-then-refill pattern. Store bathroom as optional context.

Impact and collision sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Heavy impact	Thud	Heavy dull sound from an object hitting a surface.	A book falling flat on the floor.	Impulsive; low frequency; short decay.	Thump is often muffled and body-like; bang is sharper/louder.
Deliberate strike	Knock	Sharp repeated strike on a hard surface.	Knocking on a wooden door.	Short impacts; mid frequency; often rhythmic.	Do not confuse with typing, which is faster and smaller.
Loud impact	Bang	Sudden loud impact or explosive-like event.	A door slamming shut.	Impulsive; high level; broadband; fast attack.	Use explosion only if true explosive source is known.
Multi-object collision	Crash	Chaotic collision or breaking involving multiple components.	A tray of glasses falling.	Broadband; many transients; longer decay.	Different from bang because it has multiple impacts.
Metal impact	Clang	Loud resonant metallic impact.	Hammer striking an iron bell.	Impulsive with metallic ringing tail.	Use clank for heavier dull metal collision.
Multiple small impacts	Clatter	Rapid series of short hard impacts.	Silverware tumbling out of a drawer.	Many short transients; irregular; mid/high frequency.	Crash is larger and more chaotic.
Muffled impact	Thump	Heavy muffled impact.	Someone landing on a wooden floor.	Low frequency; soft attack compared with bang.	Often body or soft-surface related.
Small mechanism	Click	Short sharp sound from a small mechanism engaging.	Pressing a ballpoint pen.	Very short; high frequency; low to medium level.	Can resemble mouse click; use specific interface label when known.
Fracture	Crack	Sharp sound from breaking or splitting.	Stepping on a dry twig.	Impulsive; high frequency; sharp transient.	Crackle is repeated irregular small cracks.
Burst	Pop	Short explosive burst.	Balloon popping or bubble wrap.	Impulsive; sharp; often high frequency.	Can resemble gunshot; avoid weapon-specific labels without context.
Flat contact	Smack or slap	Sharp flat contact between skin or flat surfaces.	A high-five between friends.	Impulsive; broadband; fast decay.	Clap usually involves two palms meeting symmetrically.
Door impact	Slam or bang	Loud impact from a door closing hard.	A door slamming in the wind.	Bang-like; often followed by room resonance.	Use source-neutral impact label unless the door source is clear from audio or metadata.
Object handling	Drop impact	Generic object falling and hitting a surface.	A cup dropped on a desk.	Impulsive; material dependent.	Use thud, clang, clatter, or crash when the acoustic evidence supports a more specific impact label.
Large energy event	Explosion	Sudden very loud broadband burst from combustion or rapid pressure release.	A firework burst or gas explosion.	Impulsive; very high level; strong low-frequency pressure wave; long decaying rumble tail.	Use only when source context (fireworks, demolition, gas event) is reliable. Use bang or loud bang if uncertain. Do not confuse with thunder, which has no sharp pressure onset.
Door mechanism	Door opening or closing	Quiet to moderate mechanical sound of a door moving on hinges and latching.	An interior door pulled gently closed.	Short; low to mid frequency; hinge creak possible; soft latch click at end.	Use slam or bang when the closing is forceful. Use creaking for the friction-only sound without the impact. Store door as

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
					optional source/context only when clearly audible.
Breakage	Glass shattering	Sharp breaking of glass into fragments, with high-frequency tinkling debris.	A dropped glass shattering on a tile floor.	Impulsive sharp onset; broadband burst; high-frequency tinkling tail.	Distinct from crash by the bright glass debris tail. Safety-relevant; label even when faint.

Friction and texture sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Hard friction	Scraping	Harsh grating from dragging a hard object across a surface.	Chair legs scraping across tile.	Noisy; mid/high frequency; sustained movement.	Grinding is usually harder, denser, and more continuous.
Fine friction	Scratching	Light abrasive sound from a pointed object on a surface.	A cat scratching a post.	Short repeated high-frequency strokes.	Can be animal or object; source may define higher-level label.
Flexible material	Scrunching or crinkling	Compressing or crumpling flexible material.	Crumpling a paper bag.	Irregular crackles; high frequency; dry texture.	Use page turning if the action is clearly pages moving.
Tight friction	Squeaking	High-pitched short friction sound.	Wet shoe on a polished floor.	High pitch; narrowband; short or repeated.	Squealing is longer and more intense.
Hard surfaces	Grinding	Harsh continuous friction from hard surfaces rubbing.	Coffee grinder crushing beans.	Dense rough noise; mid/high energy; sustained.	Appliance grinding may include motor noise.
Structural friction	Creaking	Slow strained sound from wood, metal, or joints under pressure.	Old door swinging open.	Low to mid pitch; slow onset; intermittent.	Structural groaning is deeper and longer.
High friction	Squealing	Prolonged high-pitched friction screech.	Car tires on a sharp turn.	High pitch; loud; sustained; narrowband plus noise.	Use brake screech if source is vehicle brakes.
Paper	Rustling or crinkling sound	Soft movement of paper sheets or book pages.	Turning pages in a book.	Soft high-frequency rustle; repeated brief swishes.	Avoid source-specific labels such as paper rustling or leaves rustling unless the source is clearly known from audio or verified metadata.
Abrasive material	Abrasive rubbing	Dry rough friction from abrasive surface movement.	Sanding a wooden block by hand.	Continuous rough high-frequency texture.	Store sandpaper or sanding as optional source/context only when clearly known.
Surface cleaning	Brushing or sweeping	Repeated light contact of bristles against a surface.	Sweeping crumbs from a table.	Soft repeated strokes; broadband high-frequency texture.	Can resemble leaves rustling depending on context.
Sliding mechanism	Drawer or cabinet sliding	Sliding sound of a drawer or cabinet runner opening or closing, often ending in a soft stop.	Pulling open a kitchen drawer.	Short to medium; low to mid frequency; rolling or rubbing texture; soft end thud.	Use door opening or closing for hinged doors. Label the final impact as thud or clunk if prominent.
Fastener	Zipping sound	Rapid ratchet-like sound of a zipper being opened or closed.	Zipping up a jacket or bag.	Fast dense buzz of small repeated clicks; short; high frequency.	Distinct from ratcheting tool by clothing or bag context. Use ratcheting sound for mechanical ratchets.
Cutting tool	Scissor snipping	Short metallic shearing sound of scissor blades closing through material.	Cutting paper with scissors.	Short metallic snip; high frequency; repeated; sharp transient.	Distinct from single click by the shearing slide. Store paper or fabric as optional context.

Metallic and resonant sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Metal impact	Clanging	Loud sharp metallic strike.	A fork hitting a metal pan.	Impulsive with bright ringing resonance.	Often stronger and sharper than clanking.
Small metal objects	Jangling	Light collision of small metal objects.	Keys rattling on a keychain.	Many tiny metallic transients; high frequency.	Can overlap with rattling; jangling implies metal keys/coins.
Glass/metal contact	Clinking	Light clear metal or glass contact.	Wine glasses touching in a toast.	Short high-frequency ring; clear tone.	Clanking is heavier and duller.
Heavy metal	Clanking	Heavy dull metallic collision.	Chains dragging across metal.	Lower metallic impacts; repeated; heavier body.	Use clanging for sharper resonant strikes.
Small resonances	Tinkling	Series of light high-pitched metallic or glass sounds.	Coins dropping into a jar.	Light high-frequency ringing; multiple short events.	Often softer than jangling.
Resonance	Ringing	Sustained clear vibrating tone from a resonant object.	A tuning fork after being struck.	Tonal; decaying tail; narrowband harmonics.	Phone ringing belongs under electronic alert.
Loose parts	Rattling	Rapid loose vibration of small objects or parts.	Pills shaking inside a bottle.	Repeated impacts; irregular; material dependent.	Mechanical rattle has machine context.
Bell/chime	Ringing sound	Clear resonant bell tone after strike.	A handbell ringing.	Tonal; long decay; strong harmonics.	Use source-neutral ringing if bell, phone, or metal source is uncertain.

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Metal drag	Metallic dragging	Metal chain pulled across a hard surface.	Dragging chains on concrete.	Repeated clanks plus scrape; metallic texture.	Use if dragging is metallic and foreground; otherwise use scraping or clanking.

Mechanical, appliance, and industrial sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Rotating machine	Whirring	Smooth continuous rotating machine sound.	A computer fan spinning.	Steady; mid/high frequency; often tonal.	Fan whoosh includes airflow; whirring emphasizes motor rotation.
Electrical/motor	Hum	Low steady vibration from electrical equipment.	A refrigerator running.	Low frequency; continuous; tonal or narrowband.	Use source-neutral hum unless the mechanical or appliance source is clear.
Loose machine	Rattle	Loose repetitive vibration from parts.	Loose bolt in a running engine.	Repeated irregular impacts over machine bed.	Use source-neutral rattle; store loose object, machine, or vehicle context separately if known.
Engine	Roar	Loud sustained powerful engine sound.	Jet engine at takeoff.	Broadband; high level; strong low-frequency energy.	Use source-neutral roar if engine, aircraft, or appliance source is uncertain.
Mechanical engagement	Clunk	Heavy solid mechanical engagement.	Car gear shifting into place.	Short low-frequency impact; mechanical resonance.	Can resemble thud; clunk has machine context.
Brake/friction	Screeching sound	High-pitched friction from brakes or metal on rails.	Train brakes engaging.	High pitch; loud; sustained squeal.	Use brake screech only when the brake source is clear.
Appliance	Low-frequency hum	Steady low motor vibration from compressor.	Refrigerator compressor running.	Low steady hum; long duration.	Use this instead of naming compressor or appliance if source is uncertain.
Air handling	Whooshing sound	Rushing air from fan or vent.	Hair dryer blowing at full power.	Broadband airflow; steady; high-frequency hiss.	Use source-neutral whoosh if fan, vent, vehicle, or airflow source is uncertain.
Kitchen appliance	Whirring or grinding	High-speed blender motor and blade interaction.	Blender crushing ice.	Loud motor whir plus grinding impacts.	Use this source-neutral label if the exact appliance or tool is not clear.
Kitchen appliance	Wet grinding sound	Labored motor and grinding food waste.	Garbage disposal processing scraps.	Harsh dense grinding; motor bed; variable load.	Use generic grinding if wet texture is not audible.
Laundry appliance	Rumbling sound	Deep continuous vibration during wash or spin.	Washing machine on spin cycle.	Low-frequency rumble; periodic rotation.	Use source-neutral rumble if the appliance or vehicle source is uncertain.
HVAC	Droning sound	Monotonous sustained low hum from HVAC.	Air conditioner running continuously.	Steady low hum plus airflow; long duration.	Use source-neutral drone if HVAC, engine, or room tone source is uncertain.
Cleaning appliance	Suction motor noise	Loud suction and motor noise.	Vacuum cleaning a carpet.	Broadband; motor tone; sustained airflow.	Use only if suction plus motor texture is clear; otherwise use whirring, drone, or whoosh.
Industrial ambience	Machine noise	Dense background of industrial motors and mechanical activity.	Machines operating in a workshop.	Continuous; mixed tonal and broadband components.	Use specific source only when clearly identifiable; otherwise label the dominant acoustic texture.
Clock	Ticking sound	Regular periodic mechanical pulse from a clock or timing mechanism.	A wall clock ticking in a quiet room.	Repeated isolated clicks; steady rhythm; low to medium level.	Use this for sustained periodic ticking. Use click for an isolated mechanism. Use timer or countdown beeping for electronic tones.
Electrical	Electrical buzz or vibration	Continuous low buzz or vibratory sound from electrical equipment or a vibrating device on a surface.	A phone vibrating on a desk or a dimmer switch buzzing.	Periodic low to mid frequency; rough buzzing texture; often 50 to 120 Hz harmonics.	Use hum or low-frequency hum for smooth tonal hum. Use buzzing insect only for insects. Store device or mains as optional context.

Air and pressure sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Gas release	Hiss	Steady release of pressurized gas.	Air escaping from a punctured tire.	Broadband high-frequency noise; sustained.	Animal hiss has biological source and often shorter context.
Fast airflow	Whoosh	Rushing sound of air moving quickly past.	A car speeding by.	Broadband airflow; level rises and falls.	Vehicle pass-by adds engine/tire components.
Short airflow	Puff	Short soft burst of air.	Blowing out a candle.	Short broadband burst; low to medium level.	Can be mouth-generated; source taxonomy may determine category.
Aperture tone	Whistling airflow	Tone from air forced through a narrow opening.	Wind through a cracked window.	High narrowband tone; sustained or fluctuating.	Human whistling has melodic control and source context.

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Steam/gas	Pressure-release hiss	Pressurized steam or vapor escaping.	Steam escaping from a kettle.	Hiss plus high-frequency noise; may include tonal whistle.	Use source-neutral pressure-release hiss; store steam, gas, or tire as optional context if clear.
Leak	Hissing sound	Air escaping from a tire or valve.	A punctured bicycle tire leaking.	Soft sustained hiss; high-frequency; close source.	Use source-neutral hiss if tire, gas, steam, or snake source is uncertain.
Deflation	Deflation hiss or whoosh	Air release from a balloon opening.	Letting air out of a balloon.	Rasping or squealing airflow; pitch changes over time.	Use when gradual deflation is audible; balloon source is optional context.

Electronic, alert, and interface sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Signal	Beep	Short electronic tone used as a signal.	Microwave finishing its cycle.	Short tonal event; narrowband; repeated or single.	Use notification tone if it is a recognizable digital alert.
Warning	Alarm or siren	Loud urgent warning tone, often oscillating or repetitive.	Fire alarm blaring in a building.	High level; tonal; repeated or sweeping frequency.	Vehicle siren can be transportation-specific.
Signal	Buzzer	Harsh continuous electronic tone.	Game show wrong-answer buzzer.	Rough tonal/noisy; sustained; mid frequency.	Buzzing appliance is mechanical/electrical noise.
Digital	Notification tone	Short distinctive digital sound.	Phone notification ping.	Brief tonal/melodic; clean envelope.	Avoid labeling if it is part of music or video content.
Timer	Timer or countdown beeping	Repeated ticking or beeping approaching zero.	Oven timer going off.	Regular repeated beeps; often increasing urgency.	Can overlap with alarm; timer usually lower urgency.
Door alert	Doorbell or chime tone	Tonal signal indicating someone at the door.	Ding-dong doorbell ring.	Two-tone or chime-like; decaying resonance.	Mechanical chime may also fit music/tonal.
Typing/interface	Typing or keystrokes	Rapid repetitive tapping on keyboard keys.	Fast typing on a mechanical keyboard.	Many small clicks; irregular rhythm; high frequency.	Use if repeated key-like taps are audible; do not rely on visible typing alone.
Interface	Interface click	Short precise snap from mouse button press.	Double-clicking to open a file.	Short click; repeated pairs; close microphone.	Use source-neutral click if mouse, pen, switch, or other mechanism is uncertain.
Interface	Tapping sound	Soft muted taps on glass surface.	Tapping on a smartphone screen.	Soft high-frequency taps; low level.	Use source-neutral tapping if surface or device is uncertain.
Small mechanism	Mechanical click	Click of a retractable pen mechanism.	Nervously clicking a pen.	Small sharp click; repeated; high frequency.	Use source-neutral click if mouse, pen, switch, or other mechanism is uncertain.
Switch	Switch snap or click	Crisp snap of a toggle or light switch.	Flipping a wall switch.	Short mechanical snap; medium high frequency.	Use only when switch-like action is audible or context is reliable; otherwise use click.
Telephony	Phone ringing tone	Repeated alert tone from telephone or smartphone.	Phone ringing in another room.	Tonal or melodic; repeated cadence.	Use if the sound is tonal and ringing; store phone, bell, or alarm source as optional context.
Sound reproduction	Media playback sound	Mixed reproduced audio from a television, radio, or streaming device, often combining speech, music, and effects.	A television playing in the next room.	Variable broadband; mixed speech, music, and effects; loudspeaker coloration.	Use music when reproduced content is clearly musical. Use talking or chatting only for live in-room speech, not reproduced speech.

Music and tonal sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Chime	Chime	Clear melodic resonant tone from chimes.	Wind chimes in a breeze.	Tonal; high frequency; long decay; often multiple pitches.	Doorbell can be chime-like but is an alert.
String	Strumming sound	Sweeping sound across multiple strings.	Strumming an acoustic guitar chord.	Harmonic; decaying; multiple string onsets.	Twang is a single sharp pluck.
Percussion	Drum beat	Rhythmic strike on a drum membrane.	Bass drum in a marching band.	Impulsive; low to mid resonance; rhythmic.	Use generic impact only if musical context is absent.
Resonant	Bell ringing tone	Sustained resonant tone from a bell.	A bell struck once.	Tonal; strong decay; harmonic or inharmonic partials.	Use if the sound is tonal and ringing; store phone, bell, or alarm source as optional context.
String/pluck	Twang	Sharp vibrating pluck of a taut string.	Releasing a rubber band.	Fast attack; pitch bend; decaying resonance.	May be non-musical; label based on action/context.
Keyboard instrument	Piano note	Hammered string tone from piano.	A single piano key played.	Tonal; sharp attack; decaying harmonic tail.	Use music if instrument not central to task.

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Wind instrument	Flute tone	Clear sustained tone from flute-like instrument.	Playing a flute note.	Smooth tonal; weak harmonics; sustained.	Can resemble whistle; instrument context matters.
Brass/horn	Horn or trumpet	Loud bright brass or horn tone.	Trumpet playing a note.	Tonal; strong harmonics; high energy.	Vehicle horn belongs to transportation.
Rhythm	Small percussion tone	Small struck musical object producing a tone.	Handbell in a classroom.	Impulsive onset with clear tonal decay.	Can be metallic/resonant if non-musical context.

Food preparation and cooking sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Preparation	Chopping	Rhythmic blade-on-board impacts.	Chopping vegetables on a cutting board.	Repeated sharp impacts; regular rhythm.	Dicing is faster and lighter.
Preparation	Slicing	Smooth gliding cut through food.	Slicing bread with a serrated knife.	Frictional cutting; lower impact; repeated strokes.	May include sawing texture with serrated knife.
Preparation	Dicing	Rapid light chopping into small pieces.	Dicing onions finely.	Fast small impacts; high-frequency blade contact.	Subtype of chopping.
Preparation	Stirring	Circular swishing of utensil through liquid or mixture.	Stirring batter in a bowl.	Soft repetitive swish; container contact clicks.	Whisking is faster and airier.
Preparation	Whisking	Rapid beating motion in liquid.	Whisking eggs.	Fast repeated swishes; high-frequency air/liquid texture.	Can resemble brushing if dry context absent.
Preparation	Grinding or crushing	Pulverizing solid ingredients.	Grinding spices with a mortar and pestle.	Repeated crushes; rough granular texture.	Store food, tool, or appliance context separately unless the source is clear.
Preparation	Peeling	Tearing or stripping food skin.	Peeling an orange by hand.	Soft tearing; irregular; mid/high frequency.	Can resemble paper tearing.
Preparation	Grating	Repetitive scraping against a grater.	Grating cheese over pasta.	Regular scrape strokes; high-frequency friction.	Close to scraping; kitchen context helps.
Preparation	Kneading	Rhythmic pressing and folding of dough.	Kneading bread dough.	Soft thumps; sticky texture; repeated pressure.	Can be low level and close-mic dependent.
Cooking	Boiling	Rapid bubbling of heated liquid.	Water boiling vigorously for pasta.	Dense bubbles; wet; continuous; medium level.	Simmering is gentler and less dense.
Cooking	Frying	Crackling and popping of food in hot oil.	Deep-frying chicken.	Dense sizzle with irregular pops.	Sizzling is steadier; frying may include food crackle.
Cooking	Simmering	Gentle low bubbling below boiling.	Soup simmering on low heat.	Soft bubbling; intermittent; low to medium level.	Use boiling when bubbles are rapid and vigorous.
Cooking	Steaming	Soft hiss of water vapor escaping.	Steam from a rice cooker.	Sustained hiss; soft; high-frequency airflow.	Steam release under pressure can be air/pressure.
Cooking	Popping sound	Sudden bursts from expanding kernels.	Popcorn in a microwave.	Irregular pops; increasing then decreasing rate.	Use source-neutral pop if popcorn, bubble wrap, or fire source is uncertain.
Cooking	Crackling sound	Irregular snapping from heat on food surface.	Pork skin crackling in an oven.	Small sharp crackles; high frequency; irregular.	Use source-neutral crackle if food, fire, or plastic source is uncertain.

Transportation and vehicle sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Road vehicle	Vehicle pass-by	Moving car sound passing a listener.	A car driving by on a street.	Level and spectrum rise then fall; engine plus tire noise.	Use whoosh if only air movement is audible and vehicle source is not clear.
Road vehicle	Road vehicle horn	Loud warning tone from a vehicle.	Car honking in traffic.	Tonal; high level; short or sustained.	Use horn tone if vehicle context is unclear.
Emergency vehicle	Siren	Oscillating warning tone from emergency vehicle.	Ambulance passing with siren.	Sweeping tone; high level; moving source Doppler possible.	Use generic alarm if siren pattern is not clear.
Rail	Track or vehicle rumble	Low rolling vibration of a train moving.	Train passing on tracks.	Low frequency rumble; rhythmic rail impacts.	Use rumble if train or vehicle source is uncertain.
Rail	Rail horn	Loud low/mid warning horn from train.	Train horn at a crossing.	Very loud tonal; long duration; strong low harmonics.	Use horn tone if vehicle context is unclear.
Rail	Brake screech	High-pitched rail/brake friction.	Train slowing into a station.	Sustained high screech; metallic friction.	Same acoustic family as brake screech.
Aircraft	Aircraft takeoff roar	High-power aircraft engine and airflow.	Jet taking off from a runway.	Loud broadband roar; rising intensity; low and high components.	Use roar if aircraft context is not clear or not required by the task.
Road vehicle	Engine revving	Engine acceleration from motorcycle.	Motorcycle revving at a stoplight.	Loud tonal engine harmonics; rapid pitch changes.	Use source-neutral engine revving if motorcycle, car, or tool engine is uncertain.

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Bicycle	Bell sound	Small bright bell signal from a bicycle.	Cyclist ringing a bell.	High-pitched metallic chime; short decay.	Use ringing/chime if bicycle source is not clear.
Road friction	Tire squeal	High-pitched tire friction during sharp turn or braking.	Car tires squealing in a parking lot.	Loud high pitch; sustained; frictional.	Brake screech is more metallic and brake-specific.
Aircraft	Rotor sound	Periodic blade chopping and engine sound.	Helicopter flying overhead.	Low-frequency pulses plus broadband rotor wash.	Use if periodic rotor modulation is clear; otherwise use engine/aircraft noise.
Truck alert	Reversing beep	Repetitive warning beep from reversing truck.	Truck backing up at a loading dock.	Regular electronic beeps; outdoor/vehicle context.	Use timer/countdown beep if vehicle context is not clear.

Construction and tool sounds

Subcategory	Canonical label	Description	Typical example	Acoustic profile	Confusable labels and annotation note
Hand tool	Hammering	Repeated hammer impacts on nail or surface.	Hammering a nail into wood.	Sharp repeated impacts; rhythmic; broadband.	Can resemble knocking; hammering is stronger and work-like.
Power tool	Drilling	Rotating drill motor and bit contact.	Power drill making a hole.	Motor whir plus high-frequency cutting; sustained.	Can resemble blender/motor if context absent.
Power tool	Sawing	Cutting material with hand or power saw.	Sawing a board.	Repeated friction teeth; rhythmic; rough high-frequency.	Hand saw is more periodic; power saw includes motor.
Surface finishing	Sanding	Abrasive smoothing of material.	Sanding a tabletop.	Continuous rough friction; high-frequency texture.	Powered sanding includes motor bed.
Heavy tool	Jackhammer	Rapid heavy percussive construction tool.	Breaking pavement with a jackhammer.	Very loud repeated impacts; strong low/mid energy.	Highly distinctive rhythmic machine impact.
Fastener	Staple gun	Sharp mechanical impact inserting a staple.	Staple gun fastening fabric.	Single sharp snap; medium/high frequency.	Can resemble click or small bang.
Small metal	Small metal objects dropping	Small metallic objects falling and colliding.	Screws spilling onto a table.	Multiple light metal impacts; high-frequency ringing.	Use source-neutral metallic drop if nails or screws are not clearly known.
Ratchet	Ratcheting sound	Repeated mechanical clicks from ratchet tool.	Tightening a bolt with a ratchet.	Regular clicks; metallic; hand-paced rhythm.	Use if repeated mechanical ratchet pattern is clear; otherwise use clicking or metallic clicking.

4. Common confusable sound pairs

Label A	Label B	Decision rule
Hum	Buzz	Hum is smoother and lower in frequency; buzz is rougher and often has stronger high-frequency components.
Clap	Slap or smack	Clap usually involves two palms and a crisp symmetric transient; slap is flat contact with one surface.
Crash	Clatter	Crash usually has larger objects, breakage, or many chaotic impacts; clatter is smaller repeated hard impacts.
Cough	Throat clearing	Cough is stronger and more explosive; throat clearing is shorter, lower, and deliberate.
Sizzle	Crackle	Sizzle is more continuous; crackle is more irregular and made of separated small snaps.
Rain patter	Typing	Rain is dense and spatially diffuse; typing has discrete key patterns and close mechanical clicks.
Wind howl	Animal howl	Wind is broadband airflow with resonances; animal howl has vocal harmonics and pitch contours.
Rattle	Source-specific rattle labels	Use rattle as the canonical acoustic label when the source is unclear; store machine, vehicle, or loose-object context separately if known.
Rustling sound	Source-specific rustling labels	Use rustling sound when paper, leaves, clothing, or another source cannot be determined from audio alone. Add source/context separately only when reliable.
Whistling	Air whistle	Human whistling often has intentional melody; air whistle is caused by wind through a gap and fluctuates with airflow.
Boiling	Bubbling	Boiling is heated liquid with dense rapid bubbles; bubbling is a general gas-through-liquid sound.
Brake screech	Tire squeal	Brake screech is metallic/frictional; tire squeal is rubber-road friction and often changes with vehicle motion.

5. Annotation guidelines and quality-control rules

Rule	Practical instruction
Dominant-source rule	If one sound clearly dominates the clip, assign the dominant label even if weaker background sounds exist.
Foreground/background rule	Use foreground labels for intentional target events. Store background ambience separately when possible.

Rule	Practical instruction
Uncertainty rule	If a label is not acoustically clear, mark uncertain rather than using visual context alone.
No visual leakage rule	Do not rely only on video evidence. For example, visible page turning should not force a page-turn label unless the source is acoustically clear.
Minimum audibility rule	Ignore events that are too quiet to be reliably heard by a trained listener at normal playback level.
Multi-label rule	Use multi-label annotation when two or more foreground events are clearly present, such as speech plus typing.
Granularity rule	Use the most specific reliable acoustic label. If the source is unclear, use the broader source-neutral label rather than guessing the object or scene.
Temporal boundary rule	For event detection, onset is the first audible evidence and offset is when the event decays into background. For continuous or sustained events (rain, HVAC, crowd murmur), onset is the start of the clip segment and offset is the end of the continuous region or the clip boundary, whichever comes first. Do not annotate individual transients within a dense continuous event; use the repeated-event rule instead.
Repeated-event rule	For dense repeated events such as applause or rain, annotate the whole continuous region instead of each transient.
Sensitive sounds rule	Use neutral, descriptive labels for bodily sounds and avoid adding medical or emotional diagnosis unless required by the task.
Source-neutral canonical label rule	Avoid using the source of sound in the canonical label when the source is not entirely clear. For example, use rustling sound instead of paper rustling or leaves rustling; add paper, leaves, or unknown in optional source/context.

Recommended final dataset columns clip_id, start_time, end_time, top_category, subcategory, canonical_label, optional_source_context, source_confidence, confidence, foreground_background, acoustic_profile, confusable_label, annotator_note

Confidence and field scale definitions

Use the following scales for the confidence and source_confidence columns. Annotators must select one value per field; do not leave blank.

Field	Value	Meaning
confidence	high	The canonical label is clear and unambiguous from audio alone. A second trained listener would assign the same label with high probability.
confidence	medium	The label is a reasonable choice but one or two plausible alternatives exist. The annotator made a judgment call.
confidence	low	The audio is unclear, masked, or degraded. The label is a best guess. Flag for adjudication if this rate exceeds 10% of clips.
source_confidence	confirmed	The optional_source_context is verifiable from audio or reliable metadata. Safe to use in training.
source_confidence	inferred	The source is a reasonable guess from context but not certain. Use with caution in model training; keep separate from confirmed sources.
source_confidence	unknown	The source cannot be identified from audio or metadata. Leave optional_source_context blank and set source_confidence to unknown.

Worked annotation example

The following shows all recommended columns filled in for a single foreground cough event in a noisy clip.

Field	Value
clip_id	clip_00412
start_time	1.24
end_time	1.89
top_category	Human respiratory and involuntary
subcategory	Reflex
canonical_label	Coughing
optional_source_context	adult human
source_confidence	confirmed
confidence	high
foreground_background	foreground
acoustic_profile	Short; impulsive repeated bursts; broadband; chest resonance
confusable_label	Throat clearing
annotator_note	Strong explosive onset distinguishes from throat clearing. Background crowd murmur present but not labeled as it is below audibility threshold.

Background, silence, and ambience labels

Use these labels when no foreground sound event is present, or when the clip consists entirely of ambient noise. These labels are required for clips that would otherwise be left blank, which causes training errors.

Subcategory	Canonical label	Description	Acoustic profile	Annotation note
Absence of event	Silence or near-silence	Clip with no audible acoustic event above noise floor.	Flat very low level; no spectral structure; RMS below -60 dBFS.	Do not leave blank. Assign this label and set confidence to high. Do not use for clips with audible room tone.
Ambience	Room tone	Low-level diffuse indoor ambience with no identifiable foreground event.	Continuous; low level; shaped noise; possible faint HVAC or electrical hum.	Use when the clip is not silent but contains no identifiable foreground event. Set foreground_background to background.
Ambience	Background noise	Outdoor or mixed environmental noise without a dominant foreground event.	Continuous broadband; variable level; no clear onset or offset.	Use for outdoor, street, or open-space recordings with no identifiable foreground. Set foreground_background to background.

Use these examples when annotators can hear the acoustic pattern but cannot reliably identify the object or source.

Avoid as canonical label if source is unclear	Use canonical acoustic label	Optional source/context
Paper rustling / leaves rustling	Rustling sound	paper, leaves, clothing, unknown
Tire air leak / steam release	Hissing sound or pressure-release hiss	tire, steam, gas, unknown
Door slam / object drop	Slam, bang, or drop impact	door, dropped object, unknown
Fan whoosh / vehicle whoosh	Whooshing sound	fan, vent, vehicle, unknown
Mouse click / pen click / switch click	Interface click (UI/device) or Mechanical click (physical mechanism)	mouse, pen, switch, unknown